

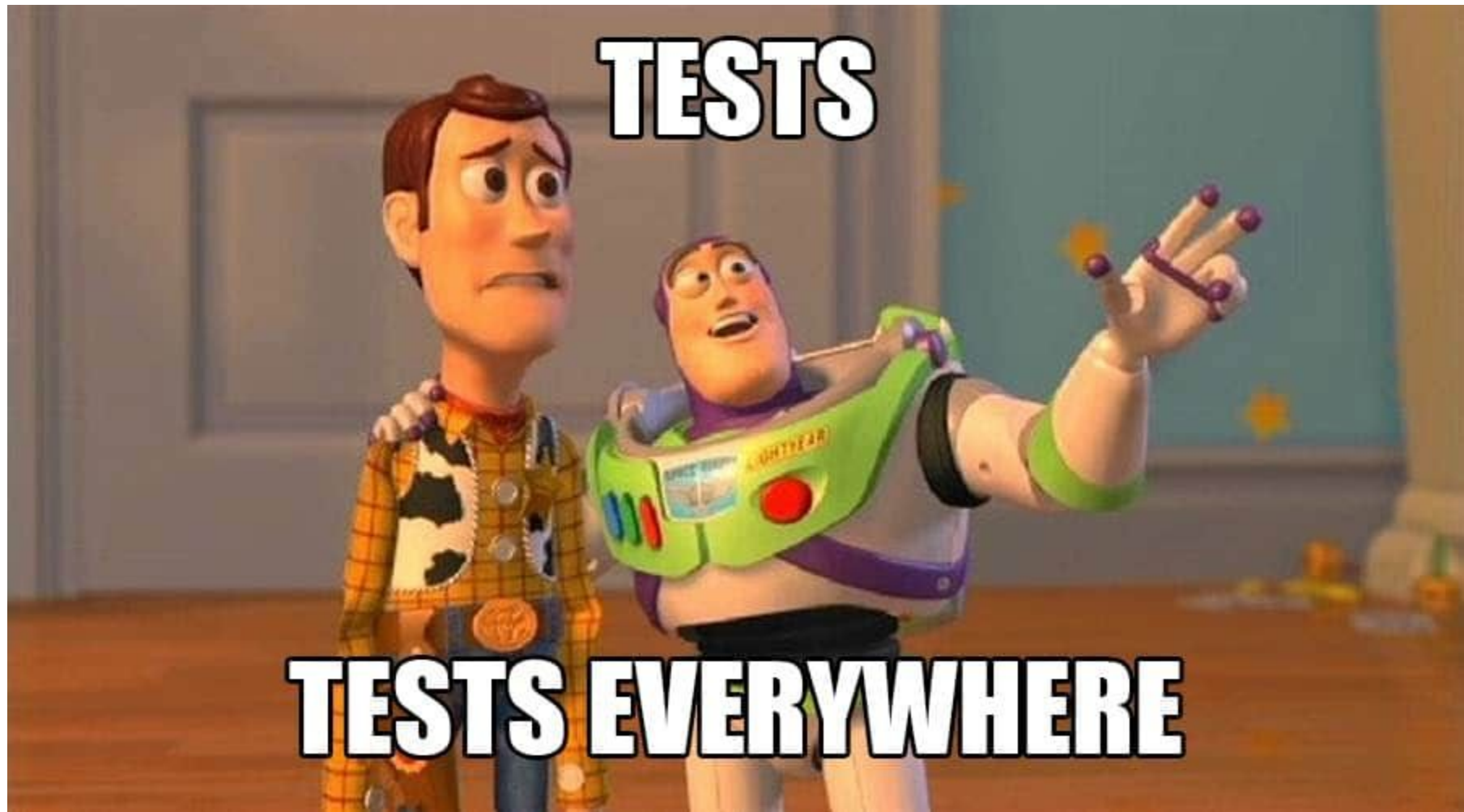
Software Testing



PREPARED BY :MITISHA BAROT

TESTS

TESTS EVERYWHERE



WHAT IS SOFTWARE TESTING?

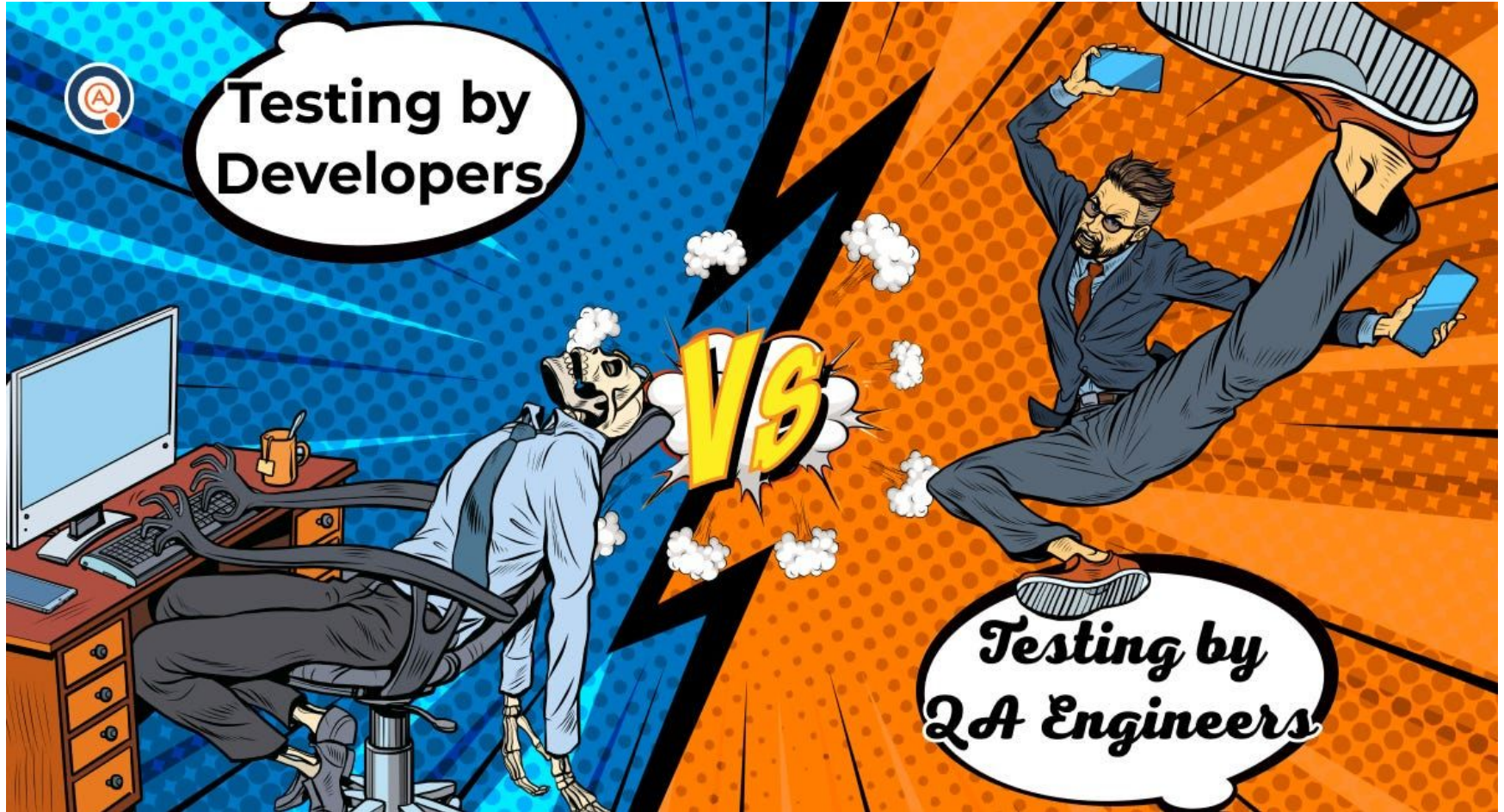




**Testing by
Developers**

VS

**Testing by
QA Engineers**



TESTER

**I FOUND A DEFECT
TODAY**

memegenerator.net

When a tester find 4 or 5 bug in software..
Tester: 😊😊



<https://softwaretestingboard.com/gta/2629>

When Developer says testing is easy

Tester :



When you are crazy about Software Testing 🤔



SOFTWARE TESTER



What my friends think I do



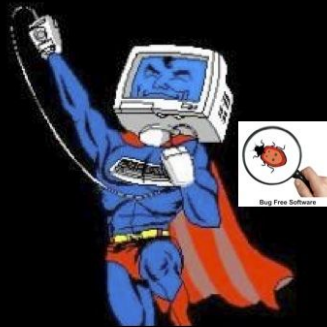
What my mom thinks I do



What society thinks I do



What programmers think I do



What I think I do



What I actually do

When a **Developer** perform Testing

QA-QC
ARENA



When a **Tester** perform Testing 🤔



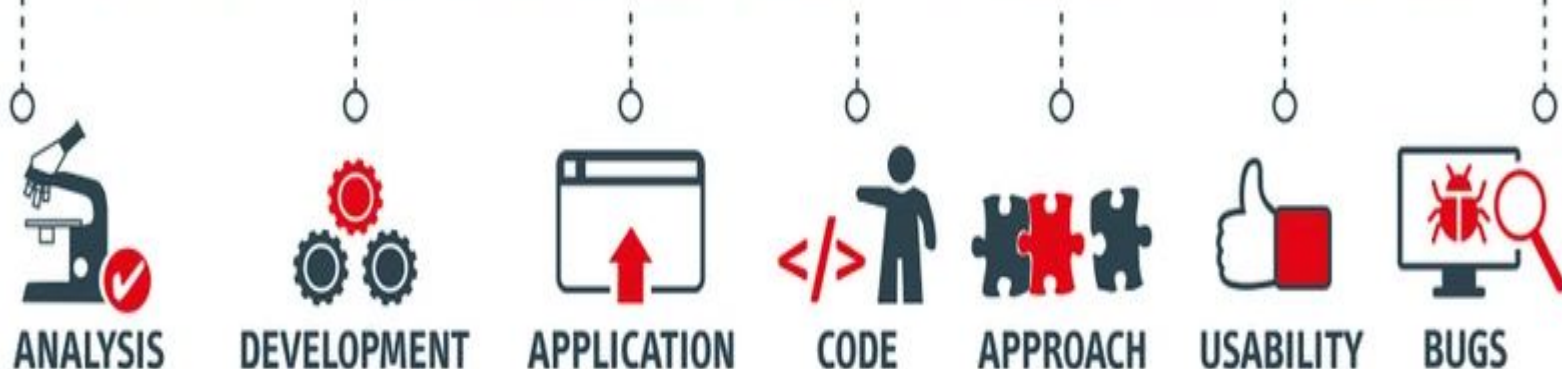
Understanding Software Testing



What is Software Testing?

The process of evaluating and verifying that a software application meets the required specifications and is free of defects.

SOFTWARE TESTING





QUALITY
LEVEL

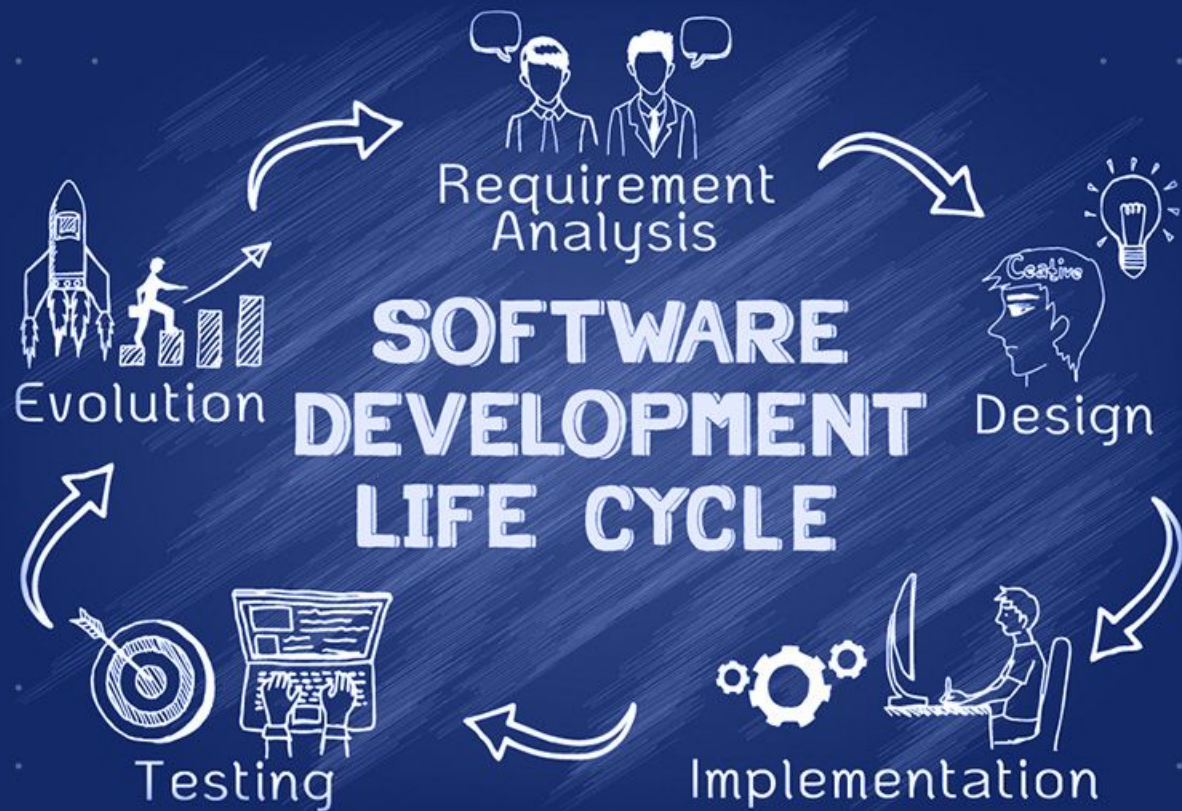


Benefits of Software Testing



6 Phases of the Software Development Life Cycle





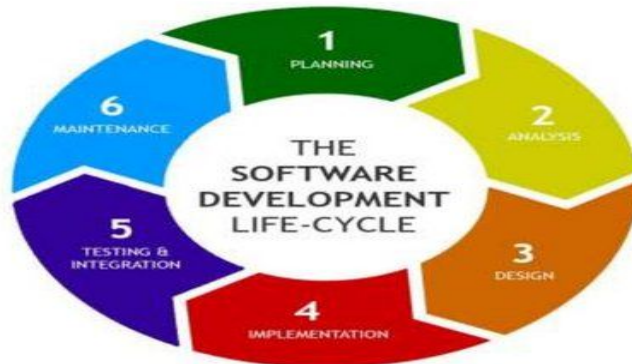
Software Development Life Cycle (SDLC) Phases

Software Development Life Cycle (SDLC) Phases



What is SDLC?

The **Software Development Life Cycle (SDLC)** is a framework defining tasks performed at each step in the software development process. **SDLC** is a structure followed by a development team within the software organization. It consists of a detailed plan describing how to develop, maintain and replace specific software.



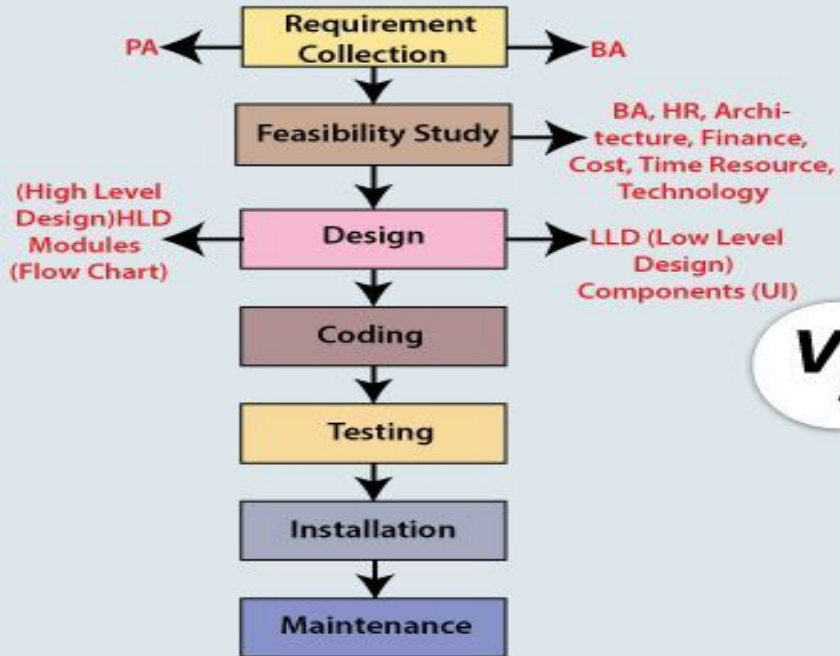


SDLC vs STLC

**SDLC****VS****STLC**

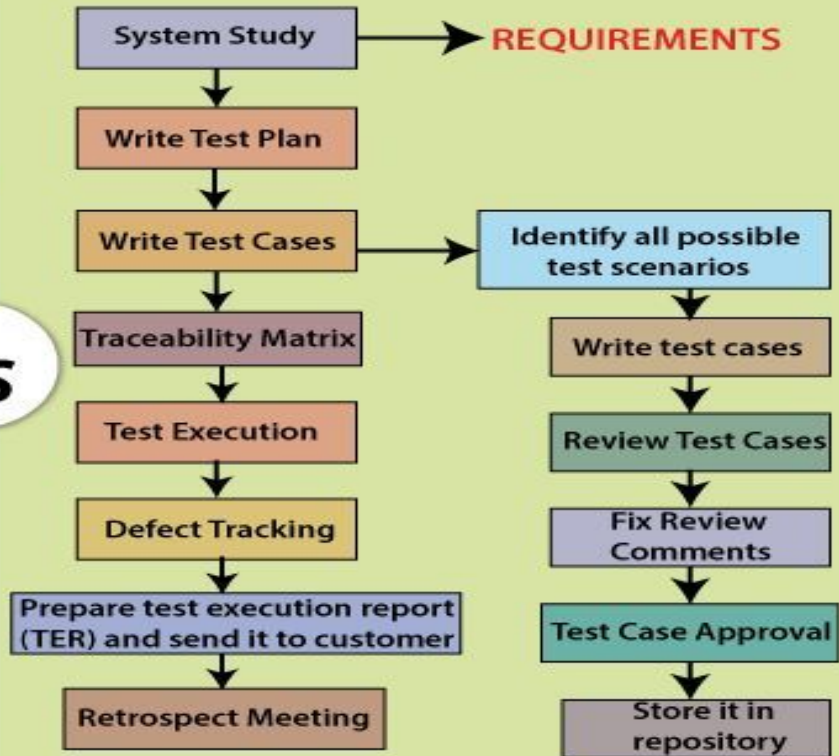
Variable	SDLC	STLC
Amplification	Software Development Life Cycle	Software Testing Life Cycle
Main objective	The main goal of SDLC is to deliver high-quality products and achieve seamless user experience through the Testing cycle	In STLC, the most important objective is to write a functional Test Plan and carry out the testing process
Planning	Coders create a well-organized Development Plan	QA team defines the Test Plan
Engineering	Devs create the actual software	Testers design Test Cases, set up the Environment, work out the RTM
Delivery	Product deployment (involves updates, post-maintenance)	QA team delivers Test results and Error metric

SDLC



Vs

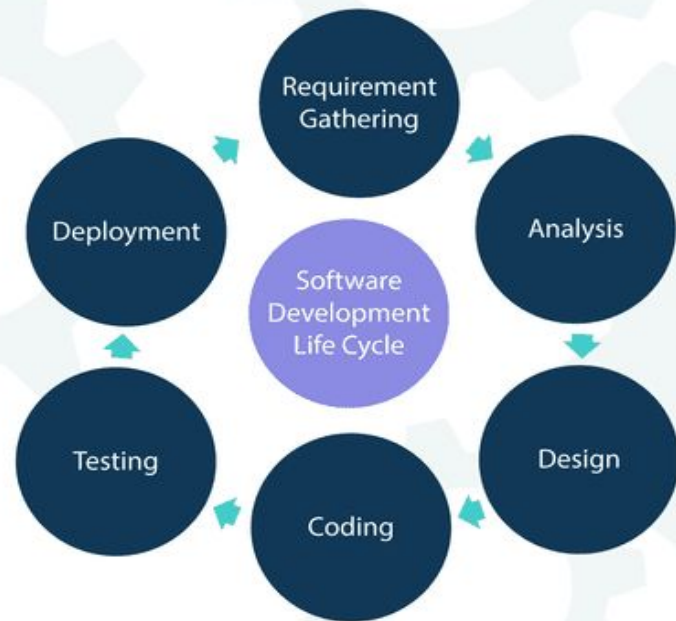
STLC



SDLC

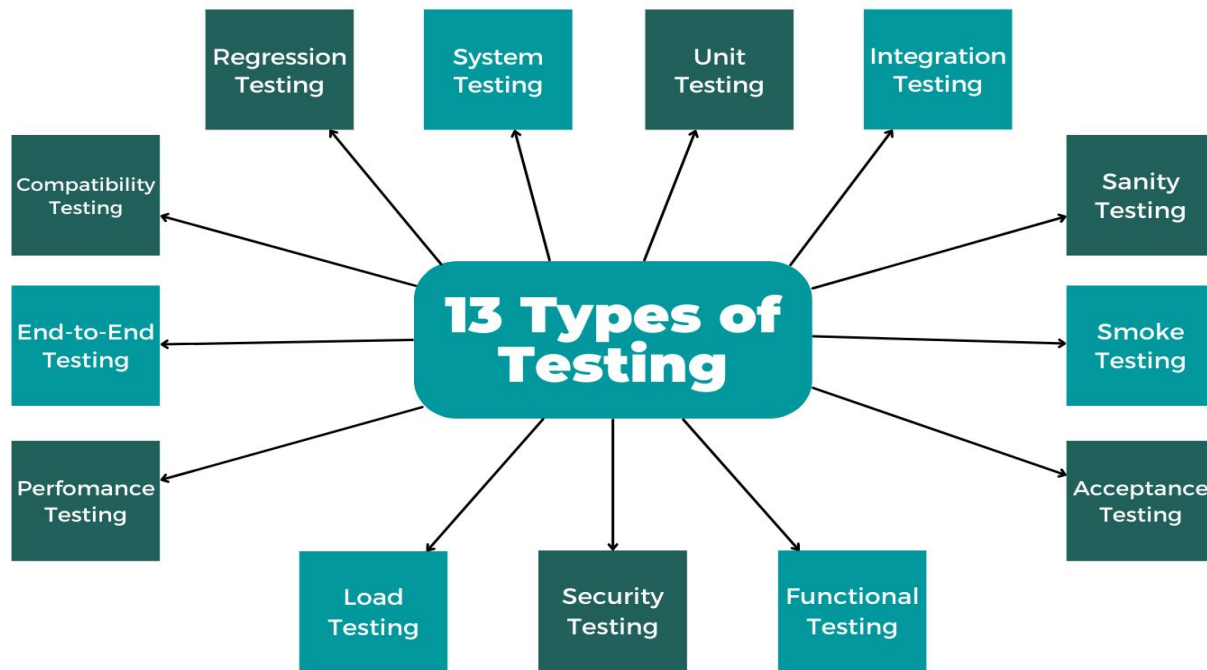
vs.

STLC



Software Testing Life Cycle





6 PHASES OF THE SOFTWARE TESTING LIFE CYCLE

1 REQUIREMENT ANALYSIS

Requirement analysis is the first step in the STLC. In this phase, software testers must understand the product requirements that need testing.

2 TEST PLANNING

Test planning successfully documents required insights into a test strategy. Teams can determine the testing time and cost in this step.

3 TEST CASE DEVELOPMENT

The test case development step creates test cases from the test plan and requirements. Test cases must cover every software functionality to be effective.

4 TEST ENVIRONMENT SETUP

Effective software testing requires a complete test environment comprising servers, test frameworks, and software tools.

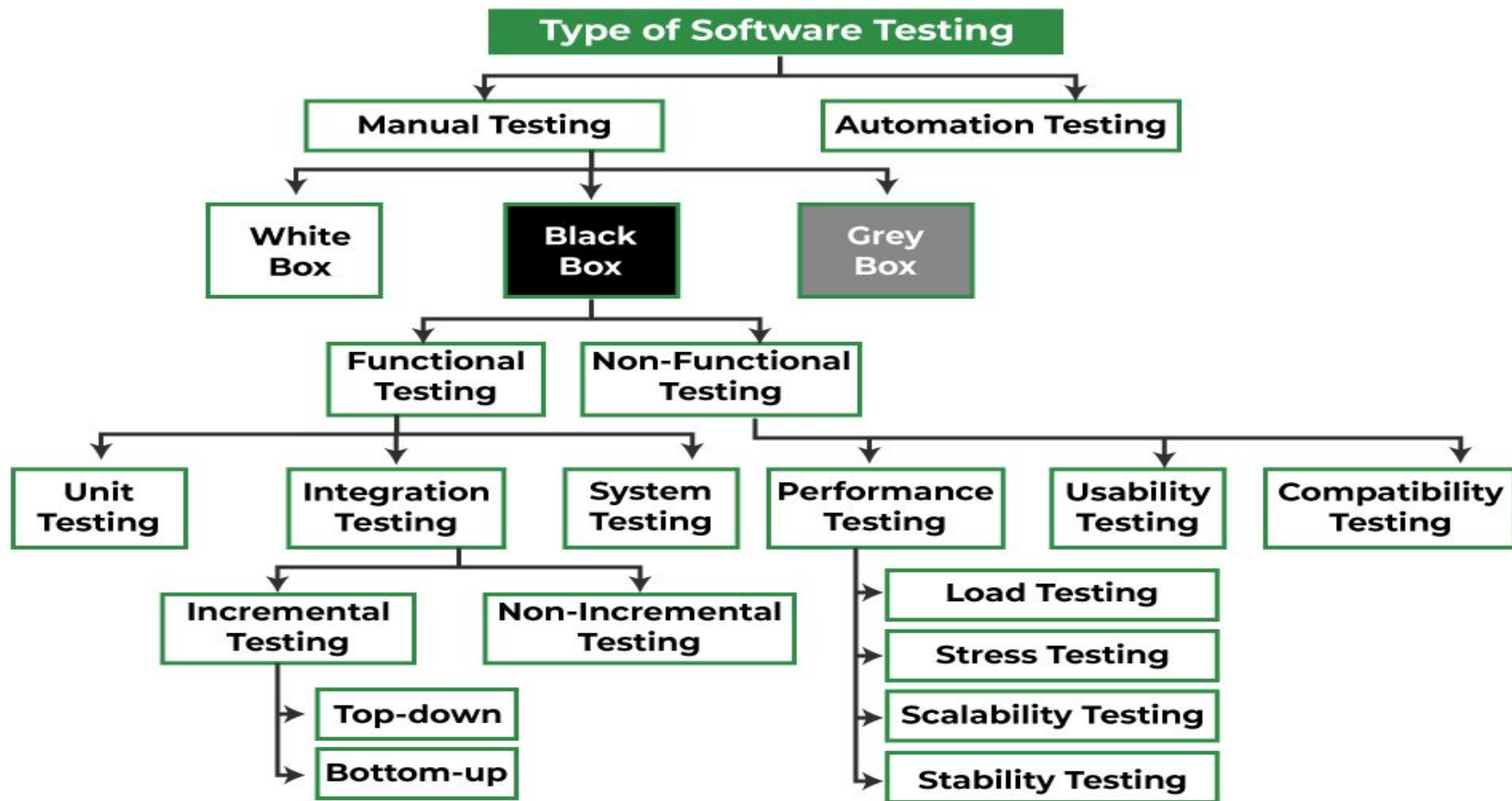
5 TEST EXECUTION

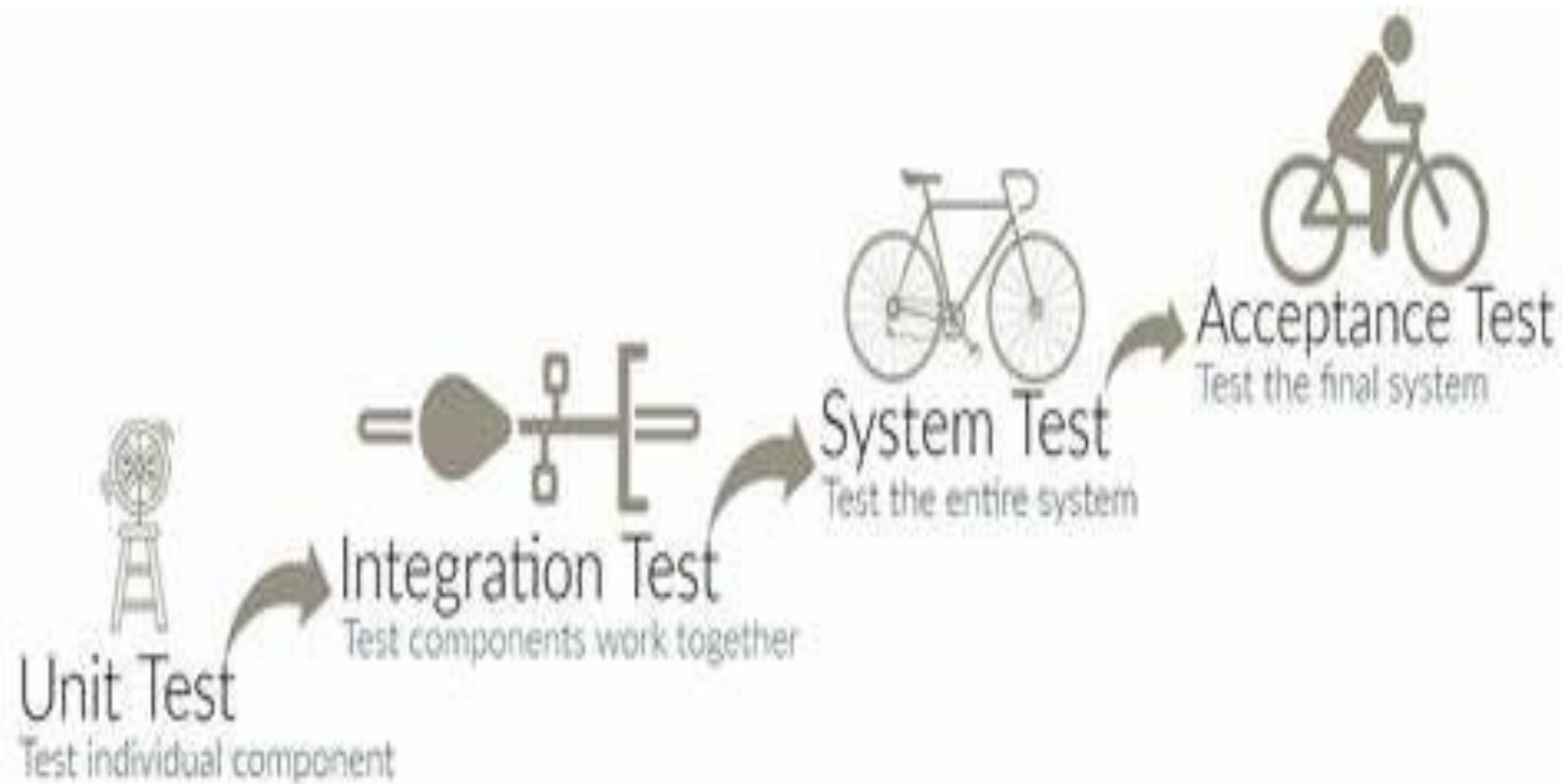
The program is ready for testing according to the test plan at this phase. This step involves running test cases to find and report software faults.

6 TEST CLOSURE

This phase concludes test case execution and product delivery.









**MANUAL
TESTING**

VS



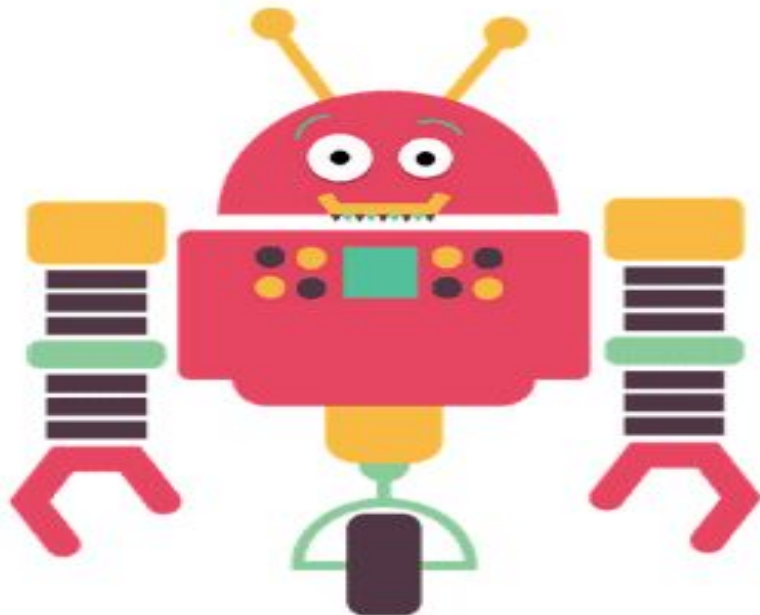
**AUTOMATION
TESTING**

Manual Testing



V/S

Automated Testing



Manual testing vs Automated testing



Manual testing



Selenium Testing Tools



```
graph LR; A((Selenium Testing Tools)) --- B[Selenium IDE]; A --- C[Selenium WebDriver]; A --- D[Selenium Grid]; A --- E[Selenium Remote Control];
```



Selenium IDE



Selenium WebDriver



Selenium Grid



Selenium Remote
Control

★ SOFTWARE REVIEW

13 Manual Testing Tools



and more...

QA process stages

Requirements analysis



Test planning



Test execution and defect reporting



Test design



Retesting and testing



Release testing



Optimization of Software Testing with AI

Improved productivity: By automating repetitive tests, AI allows QA engineers to focus on testing new features, increasing overall productivity.

Automatic test case creation: AI can automatically generate test cases, saving time and effort.

Intelligent test execution: AI can intelligently decide what and when to run tests based on code changes, optimizing test execution.

Improved test coverage: AI helps ensure comprehensive test coverage, minimizing the risk of undetected issues.

Fail-safe of Selenium tests: AI can enhance the reliability of Selenium tests, reducing false positives and false negatives.

Faster test script automation: AI-assisted co-pilot and pair programming techniques accelerate the automation of test scripts.

Automated API test generation: AI can generate automated API tests, ensuring comprehensive API coverage.

Quick error detection: AI algorithms can swiftly identify errors, enabling prompt resolution.

Visual testing capabilities: AI-powered visual testing enables the identification of UI and layout inconsistencies.

Enhanced regression testing: AI aids in performing efficient and effective regression tests, saving time and effort.

5 Stages of Manual Testing

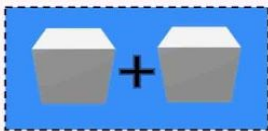
1-

Unit Testing



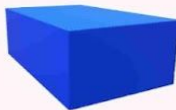
2-

Integration Testing



3-

System Testing



4-

UI Testing



5-

Acceptance Testing



For Example...

Any web application:
Check Login button functionality

ID at a platform:
Does user profile shows their chosen category content?

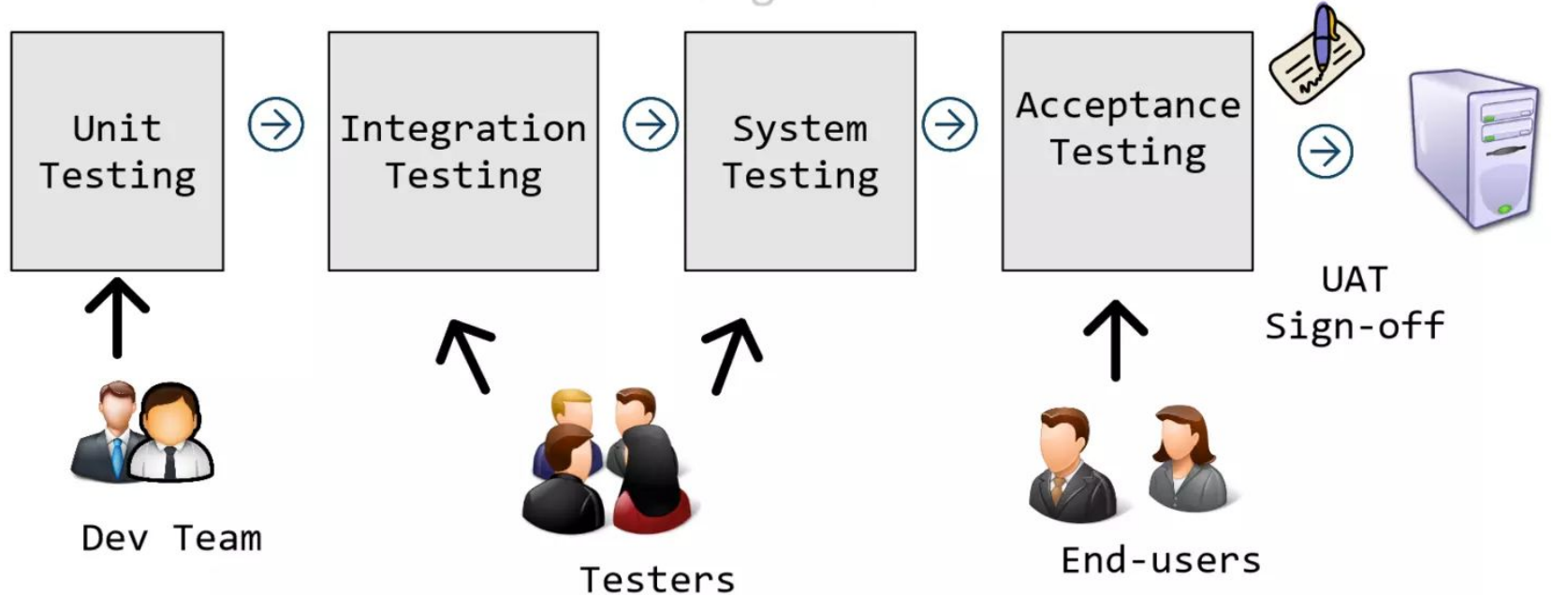
Is software as a whole providing desired output?

Check visual appeal: Color, font, box size etc

Use software in real life and measure user experience

Software Testing Levels

www.TestingDocs.com



ALREADY REGISTERED?

Email address

Password

[Forgot your password?](#)



Sign in

*check Login
Functionality*

P = Plain Text | C = Cipher Text | K = Key

Test Case ID	Test Case Objective	Pre requisite	Steps	Input Data	Expected Output	Actual Output	Status
TC_01	Test Caesar Cipher Algorithm (For Encryption)	Textfield should be enabled	1. Select Encrypt Button 2. Enter Plain Text 3. Enter Numeric Key 4. Submit	P: hello K: 3	khoor	khoor	PASS
TC_02	Test Caesar Cipher Algorithm (For Decryption)	Textfield should be enabled	1. Select Decrypt Button 2. Enter Cipher Text 3. Enter Numeric Key 4. Submit	C: khoor K: 3	hello	hello	PASS
TC_03	Test Vignere Cipher	Text Fields should be enabled	1. Enter Plain Text 2. Enter String Key 3. Submit	P: hello K: abcds	hfnog	fhgon	FAIL
TC_04	Test MD5	Text Fields should be enabled	1. Enter Plain Text 2. Submit	T: hello	5d41402a bc4b2a76 b9719d91 1017c592	5d41402a bc4b2a76 b9719d91 1017c592	PASS
TC_05	Test Columner Cipher	Text Fields should be enabled	1. Enter Plain Text 2. Enter Numeric Key 3. Submit	P: hello K: 3	hleol	hleol	PASS



KEEP
CALM
AND
CONTINUE
TESTING

